

EPHEMERIS

A generator of Berlin School music

Version 0.1.0 · 432 parameters · manual built 25.09.2026

The EPHEMERIS interface is a dark-themed digital audio workstation (DAW) for generating Berlin School music. It features a top control bar with a title 'EPHEMERIS', a 'Cosmic' dropdown menu, and a 'A' dropdown menu. Below these are buttons for 'reroll form', 'reroll tempo', 'reroll rows', 'reroll rack', 'reroll layers', 'reroll lead', 'reroll pads', 'reroll hands', 'Muted (env)', 'Save set', 'Load set', and 'Export WAV + MIDI'. The main area displays a timeline with a 'seed 4242' and a '16.0 min' duration. The timeline is divided into sections: 'Atmo', 'Einsatz', 'Aufbau', 'Aufbau', 'Aufbau', 'Lead', 'Hoehepunkt', and 'Ausklang'. Below the timeline is a 'Mixer' section with tabs for 'Perform', 'Rack', 'Gestures', 'Voices', 'Lead', 'Drone', 'Tape Keys', 'Strings', 'Atmosphere', 'Echo + Spring', 'Hall', 'Drums', 'Master', and 'Style'. The mixer contains 14 columns, each representing a different sound source: Row 1, Row 2, Row 3, Row 4, Row 5, Row 6, Row 7, Row 8, Lead, Drone, Tape Keys, Strings, Drums, and Atmosphere. Each column has a set of controls: a 'Pan' knob, an 'Echo' knob, an 'Echo 2' knob, a 'Blend' knob, and a 'Hall' knob. Below these knobs are vertical sliders for volume, with values ranging from -8.9 to -52.5. The interface is designed to be intuitive and visually appealing, with a focus on the unique sound palette of Berlin School music.

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About

Ephemeris composes and plays long pieces in the manner of the modern Berlin School: sequencer lines of different lengths turning against each other over a common root, a bass that holds the ground, tape choirs and string machines, a lead that sings over the peak, the room of a tape echo and a long hall. There are no samples. Every sound is synthesised by the engine, and every note is decided by the composer.

The same seed always gives the same piece. A piece is a seed, a style, a length and a handful of rerolls, which is what a set file stores and what you can hand to someone else. The standalone and the plugin play what the offline renderer plays, sample for sample.

An ephemeris is the table that gives the positions of the heavenly bodies for every date. A sequencer is such a table, and the music here is made of orbits of different lengths around one root that meet again only now and then. The Rack page draws them that way.

The top bar

Style, key and scale choose the piece's character: Cosmic, Doom, Melodic, Modern and Drift are five style profiles with their own tempi, forms, layers and spaces. Piece min is the length of one piece; Concert min above zero strings pieces together into a concert that travels through related keys. With Morph To (on the Master page) a concert also wanders from its style to another: each piece is composed between the two, the further into the concert the nearer the second. Concert Arc gives a concert an arc of tension: its first and last pieces quieter -- fewer rows at the peak, the lead and the drums rarer, the hands slower, the sound a little darker -- and the pieces around 60 % of the way the densest. Compose writes a piece with the current seed, New seed draws another one, Play starts and stops the standalone (in a host the host's transport plays).

Scale offers eight modes: Aeolian, Dorian, Phrygian, Harmonic Minor, Minor Pentatonic, Mixolydian, Lydian and Locrian. The harmony is the Berlin School's: a centre held by the drone and the bass, chords that change rarely. From the first build on, a chord track moves the bass row alone under the unchanged sequence -- the counter rows are heard anew over the wandering bass -- in one of four ways the style draws: one chord for the whole phase, a pendulum between two (i-VII, i-VI, i-iv, i-IV in Dorian, i-bII in Phrygian), a loop of three or four (i-VI-VII, i-VII-VI, i-iv-VI ...) or a walk by a table of chances. Drift and Modern stay longest on one chord, Melodic changes every four to eight bars. The tape keys and the strings build their chords on the same degree: the choirs mostly plain triads, the strings open fifths, triads, sus chords, add9, stacked fourths and m9 as well. The transposer moves the whole sequence a fifth up, a fourth up or a minor third down and back, only on the plateau and at the peak, while the drone stays on the centre. At some peaks Aeolian brightens to Dorian, in some breakdowns it darkens to Phrygian, and every piece ends on the open fifth. A later phase may move its key by a fourth, a tone or a third, and take a minor mode of its own, as the parts of a suite do.

The reroll buttons draw one part of the piece again and leave everything else as it was, to the note: the form, the tempo, the rows' shapes, the rack's patterns, the layers, the lead, the chords and pads, and the hands on the knobs. Save set and Load set keep a piece as a small text file; Export writes the piece offline as a 24-bit WAV and a MIDI file beside it, with the gestures as controllers -- and, if you choose so, the stems: a WAV per channel strip and one for the rooms in a folder beside the mix, whose sum is the mix before the master.

Below, the arrange view shows the sections of the piece -- the atmosphere at the start, the entry of the rows, the builds, the lead, the peak, the bridges, the fade -- with the playhead. Click to jump.

Mixer



A channel strip per source, in the order the engine mixes them: the eight rows, the lead, the drone, the tape keys, the string machine, the drums and the atmosphere. Each strip has its pan (where the source has one), its sends to the tape echo and to the hall -- the rows also to the second echo -- its fader and its meter. The controls are the same parameters the sources' own pages show.

The composer sets the stereo field for each piece: the bass in the middle, the main sequence near it, the other counter rows 30 to 50 % out to alternating sides, the lead a little off the middle. It keeps the bass dry and the tape keys and strings out of the echo, and sends the counter rows after the first into the second echo, so each layer of the sequence has a delay time of its own.

Every source has a Low Cut on its own page: a gentle high pass (12 dB per octave) that keeps it out of the bands below its own, after the production guide's frequency plan. Only the bass row reaches down to 30 Hz; the composer starts the main sequence at 90 Hz and the other counter rows at 200 Hz; the lead and the tape keys start at 150 Hz, the strings at 200 Hz, the atmosphere at 120 Hz, the drone at 40 Hz (its lowest note is at 82 Hz), the drums at 25 Hz. Mud between 200 and 500 Hz and a crowded bass are what make a long piece tiring; this is where they are kept away.

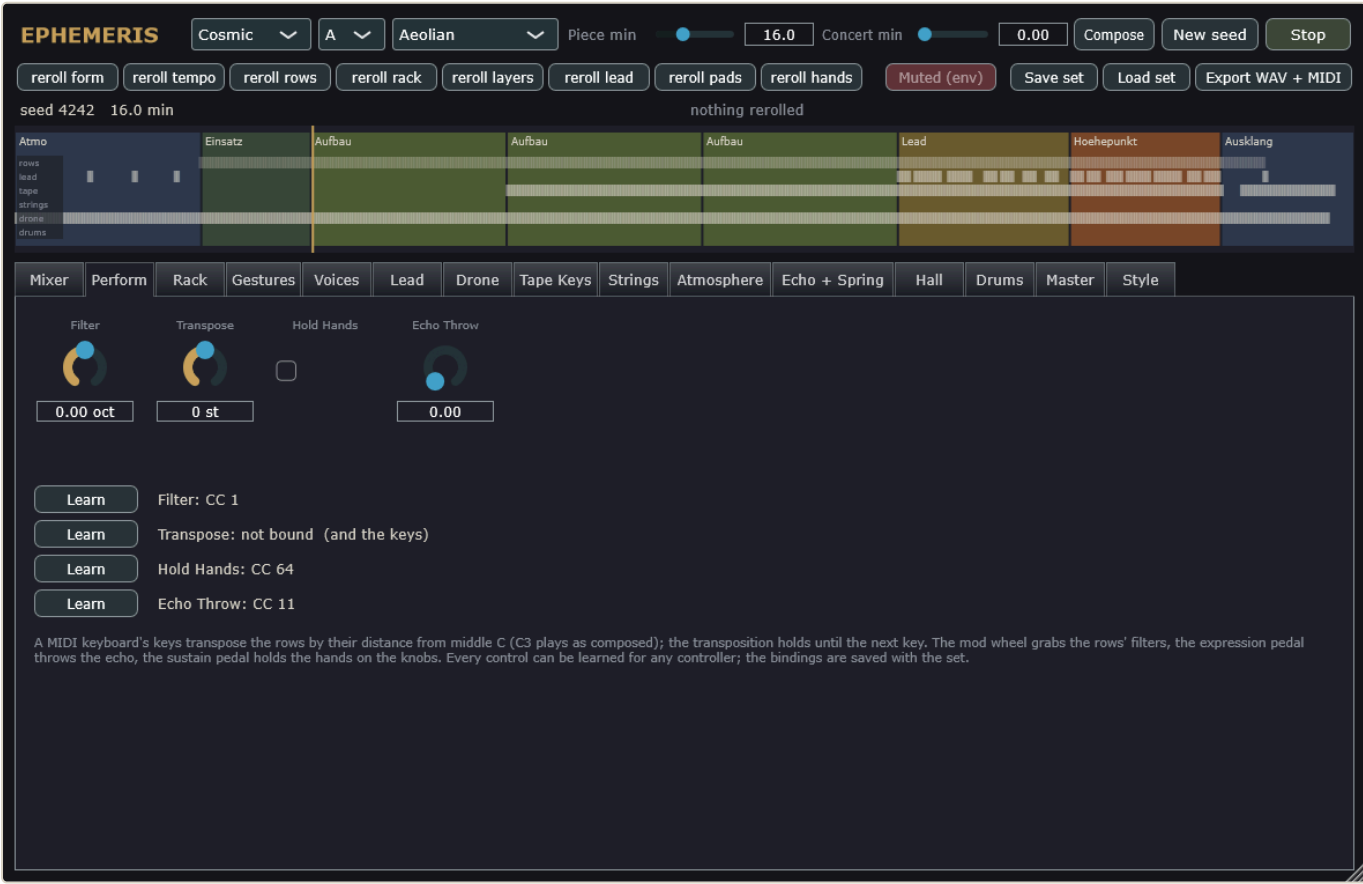
Punch (on the rows) lifts a row's attacks by up to 6 dB and leaves its sustain: the composer gives it to the main sequence, which keeps it in front. Distance (on the rows, the lead, the drone, the tape keys and the strings) places a source between near (0) and the horizon (1) with every cue at once, so nothing is near in its level and far in the hall: at 0.3 it is 4 dB quieter, its highs end at 10 kHz and it sends a little more to the hall; at 0.6 it is 10 dB back, dark from 5 kHz and well in the hall; at 1 it is 24 dB back, 2.5 kHz and almost only hall. The composer

moves the bass with it: at each entry it comes in from 0.7 over 45 seconds, in the breakdown it recedes over up to two minutes, and in the coda before it stops -- an approach and a leaving instead of a fader ride.

The rows also make room for themselves: while they play, the pads (strings and tape keys) step back in their middle band, 300 Hz to 5 kHz, by up to Cascade Duck (Master page, 2.5 dB), and the pads in turn push the atmosphere back in the same band -- the pads stay full while the sequence rests.

The meter reads what the source puts into the mix after its fader and pan: a green bar for the RMS level with a release of 300 ms, turning amber above -6 dBFS and red above 0, and a peak line that holds for a second and a half before it falls at 20 dB per second. The figure under the meter is the held peak. A strip that shows -inf is not playing at the moment.

Perform



The performer's controls act on the piece as it plays, and at their defaults they do nothing at all: the piece is then exactly as composed. Filter moves the rows' filters up to two octaves either way, as a hand on the cutoff knobs of a modular would. Transpose is the transposition key: every note that starts after a change comes in transposed; notes that sound keep their pitch. Hold Hands lets go of the knobs: the gestures the composer wrote stand where they are until you release it. Echo Throw sends every source into the tape echo and raises the echo's feedback with it, for a dub throw at the end of a phrase.

A MIDI keyboard plays the transposition key directly: a key transposes by its distance from middle C and holds until the next key, as the transposer of a sequencer did. The mod wheel grabs the filters, the expression pedal throws the echo, the sustain pedal holds the hands. Learn binds any controller to any of the four; the bindings are saved with the plugin's state.

A row cannot be mutated live: the piece is composed before it plays. To hear the rack with other patterns, reroll the rack.

PERFORM

Control	Key	Range	Default
Filter	perform.filter	-2 .. 2 oct	0 oct
Transpose	perform.transpose	-12 .. 12 st	0 st
Hold Hands	perform.hold	off / on	off
Echo Throw	perform.throw	0 .. 1	0

Rack



On the left, the orrery. Every row that plays notes is a planet on an orbit, the bass innermost; one turn is one cycle of the row, its length in steps times its step. The top of every orbit is the row's first step. When running rows stand there together, a line of light rises from the centre: a conjunction, the moment the patterns come round together. Under the orbits, each row's cycle and the time until all running rows meet again. The sun in the middle is the transposer: its letter is the root the rows play on at the moment, and it moves when the transposer does.

The composer uses the conjunctions: a build, which brings in a new counter row, ends where that row and the bass row begin their cycles together again, if such a bar is near.

On the right, the rows' parameters: length and step, direction, octave and transposition, the chance of a mutation at the end of each cycle (a Turing machine's shift register), the gate, level, pan and sends, and the mode -- notes, or the transposer that moves the others. The composer sets the rows for each piece; the values here are what a piece starts from.

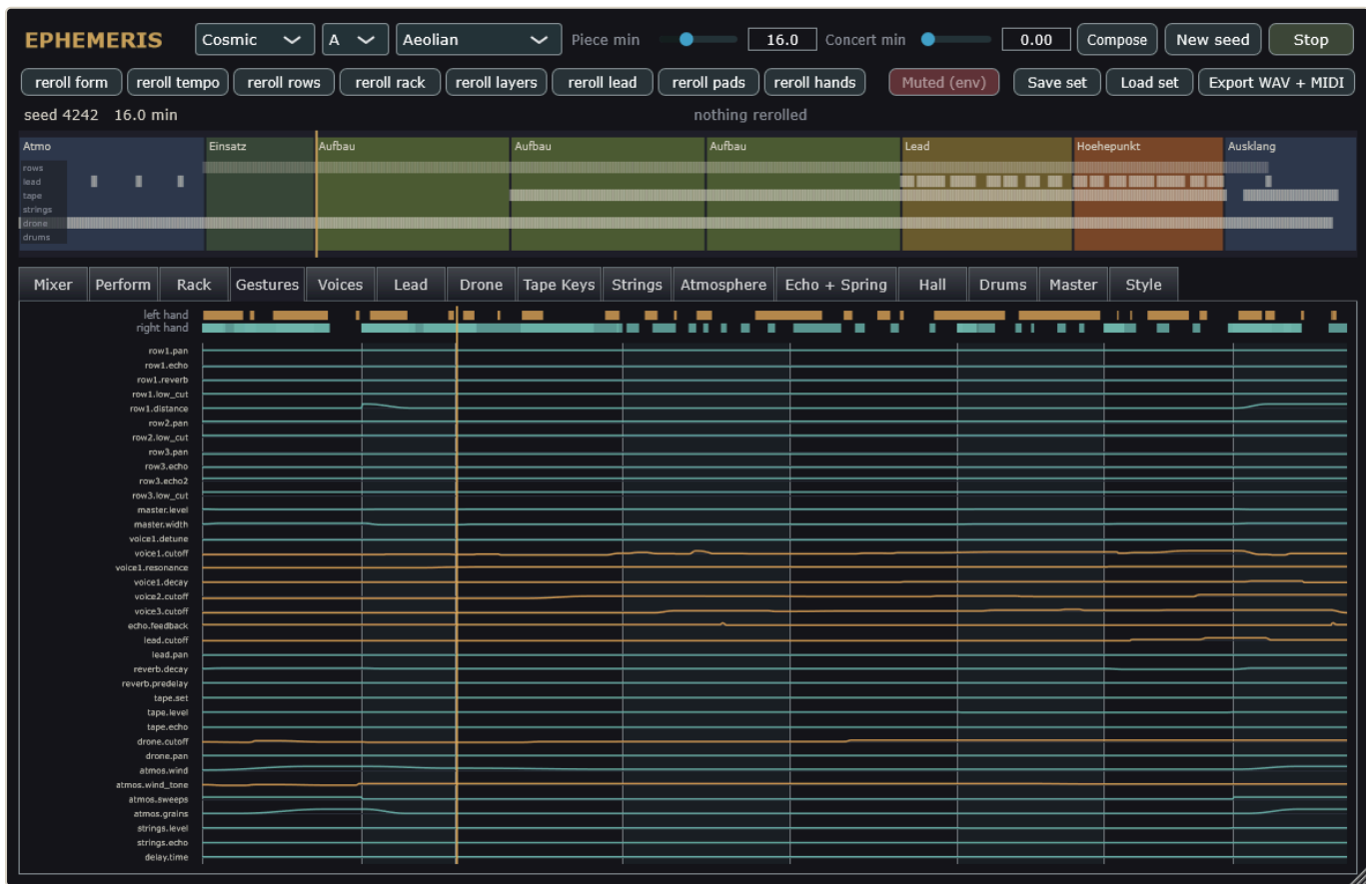
The composer draws each row from one of the Berlin School's figures: the octave pendulum, the fifth anchor, the 3+1 pulse (three notes and a rest), stairs up and down, the mode's colour (the note that makes Dorian or Phrygian heard), the Phrygian push, the arpeggio spiral from i to VI, or a canon of the row before -- three steps later or a fifth up -- besides the older free rules. The bass ostinato has eight or sixteen steps and a long gate; every eight steps have one to three rests (the bass none to two), the first always sounds on the root or the fifth, accents fall on the first and fifth step or 3+3+2. Some steps are probability gates that sound on some rounds only -- Modern has the most, Cosmic and Doom the fewest; a few steps, the most in Drift and Modern, play a new note each time round, drawn through a quantiser from the root, fifth, octave, seventh, fourth and

third of the mode. On the plateau and at the peak a few steps are ratcheted into quick triggers, at some peaks the eighths row doubles its pulse, and in the breakdown and the coda the bass loses its steps one by one.

ROW (8 INSTANCES)

Control	Key	Range	Default
Active	rowN.active	off / on	on
Length	rowN.length	1 .. 32 steps	16 steps
Division	rowN.division	1/4, 1/8, 1/8 T, 1/16, 1/16 T, 1/32, 1 Bar, 2 Bars, 4 Bars	1/16
Direction	rowN.direction	Forward, Backward, Pendulum, Random Walk	Forward
Octave	rowN.octave	-3 .. 3	-1
Transpose	rowN.transpose	-12 .. 12 st	0 st
Mutation	rowN.mutation	0 .. 1	0
Gate	rowN.gate	5 .. 100 %	50 %
Level	rowN.level	-60 .. 6 dB	-6 dB
Pan	rowN.pan	-1 .. 1	0
Echo Send	rowN.echo	0 .. 1	0.3
Mode	rowN.mode	Notes, Transposer	Notes
Reverb Send	rowN.reverb	0 .. 1	0.3
Echo 2 Send	rowN.echo2	0 .. 1	0
Low Cut	rowN.low_cut	10 .. 500 Hz	30 Hz
Distance	rowN.distance	0 .. 1	0
Blend Send	rowN.blend	0 .. 1	0
Punch	rowN.punch	0 .. 1	0

Gestures



What the player's two hands do to the knobs over the whole piece. Every knob the composer moves has a lane: its offset from where you set the knob, as a curve over the piece, from -1 to +1 of the knob's range -- the filter of the bass row opening towards the peak and closing in the coda, the counter rows' filters, the lead's, the echo's feedback thrown now and then, the wind rising and falling in the atmosphere. The colour is the hand that moves the knob.

Above the lanes, a bar per hand shows where it is busy. A player has two hands, and the composer keeps to that: never more than two knobs move at once, and a hand finishes one move before it starts the next. The sections of the piece lie behind the lanes, the playhead runs over them, and a click jumps there. The hands can be let go on the Perform page (Hold Hands): every knob then stays where the curve has it.

Voices



Each row plays its own modular voice: two drifting oscillators (a sawtooth blended into a pulse, the second detuned), a mixer driven into a soft saturation, a four-pole transistor ladder at twice the sample rate, two envelopes and glide. The oscillators run freely and are never reset on a note, and each wanders slowly in pitch, as analogue oscillators do; the Drift knob sets how far.

Presets. Every synth page -- the voices, the lead, the drone, the tape keys, the strings, the drums and the atmosphere -- has 1024 factory presets above its knobs, in sixteen groups of sixty-four ("Ladder Bass", "Glass Arp", "Cathedral Choir", "Slow Swell", "Solar Wind" ...); the arrows step through them. Each name is two words: the first runs from dark to bright ("Umbral" ... "Radiant"), the second is the group's own, so "Umbral Keel" is the darkest and "Radiant Piston" the brightest of the ladder basses. On the voices page a preset goes to the row chosen on the left. Save... keeps the synth's knobs as they stand under a name of your own (a small text file in the application data, Ephemelis/Presets); they appear under "User". A preset sets the sound, never the mix: level, pan, the sends, the low cut, the distance and the drone's auto pan stay, and so do the atmosphere's amounts, which the composer moves. Nor does it put a synth out of tune with the rest: the drone and the bass groups keep their oscillators within a cent or three of each other (the pure foundation), the others within twelve, the lead's vibrato and the tapes' wobble stay inside their usual bounds, and no preset transposes anything.

The knobs the player's hands move during a piece -- cutoff, resonance, envelope amount, decay -- are moved by the composer's gestures as offsets from where you set them, so the ranges you choose keep their meaning.

VOICE (8 INSTANCES)

Control	Key	Range	Default
Wave	voiceN.wave	0 .. 1	0
Detune	voiceN.detune	0 .. 30 ct	7 ct
Pulse Width	voiceN.pw	0.05 .. 0.5	0.5
Drift	voiceN.drift	0 .. 15 ct	3 ct
Drive	voiceN.drive	0 .. 24 dB	6 dB
Cutoff	voiceN.cutoff	30 .. 16000 Hz	650 Hz
Resonance	voiceN.resonance	0 .. 1	0.4
Env Amount	voiceN.env_amount	0 .. 6 oct	3.5 oct
Filter Decay	voiceN.decay	15 .. 2000 ms	190 ms
Key Track	voiceN.keytrack	0 .. 1	0.5
Accent	voiceN.accent	0 .. 1	0.5
Amp Release	voiceN.amp_decay	5 .. 2000 ms	60 ms
Glide	voiceN.glide	1 .. 500 ms	60 ms

Lead



The lead is the same voice as a row's, with vibrato that comes in over the first half second of a held note, as a hand reaches the wheel. It plays in its own section and over the peak: phrases of four to eight bars with rests of two to four between them, on the pentatonic of the moment (with the sixth where the mode is Dorian), chord tones on the strong beats, glides from the tone below, repeats of its own phrases, and an end on the fifth or the minor third rather than the root. It plays a little behind the beat, as a player over a machine does.

LEAD

Control	Key	Range	Default
Wave	lead.wave	0 .. 1	0.25
Detune	lead.detune	0 .. 30 ct	4 ct
Pulse Width	lead.pw	0.05 .. 0.5	0.45
Drift	lead.drift	0 .. 15 ct	2 ct
Drive	lead.drive	0 .. 24 dB	8 dB
Cutoff	lead.cutoff	30 .. 16000 Hz	1500 Hz
Resonance	lead.resonance	0 .. 1	0.25
Env Amount	lead.env_amount	0 .. 6 oct	2 oct
Filter Decay	lead.decay	15 .. 2000 ms	450 ms
Key Track	lead.keytrack	0 .. 1	0.6
Accent	lead.accent	0 .. 1	0.3

Amp Release	lead.amp_decay	5 .. 2000 ms	300 ms
Glide	lead.glide	1 .. 500 ms	80 ms
Level	lead.level	-60 .. 6 dB	-9 dB
Pan	lead.pan	-1 .. 1	0.15
Echo Send	lead.echo	0 .. 1	0.45
Vibrato	lead.vibrato	0 .. 100 ct	18 ct
Vibrato Rate	lead.vibrato_rate	1 .. 10 Hz	5.2 Hz
Reverb Send	lead.reverb	0 .. 1	0.4
Auto Pan	lead.auto_pan	0 .. 1	0
Low Cut	lead.low_cut	10 .. 500 Hz	150 Hz
Distance	lead.distance	0 .. 1	0
Blend Send	lead.blend	0 .. 1	0

Drone



The drone is the ground under the whole piece: the root, low, held from the start to the fading coda, following the key when a phase changes it. It is a modular voice as well and has the lead's table. Auto Pan lets it wander slowly across the stereo field, on a sine of 0.05 Hz (twenty seconds from one side to the other and back); the lead has the knob as well, at zero. The drone's two oscillators are not detuned against each other, nor are the bass row's: the foundation of a piece stays in pure octaves and fifths, without a slow cloud of beating in the middle over the minutes; the detune belongs to the counter rows, the lead and the pads.

DRONE

Control	Key	Range	Default
Wave	drone.wave	0 .. 1	0
Detune	drone.detune	0 .. 30 ct	0 ct
Pulse Width	drone.pw	0.05 .. 0.5	0.5
Drift	drone.drift	0 .. 15 ct	4 ct
Drive	drone.drive	0 .. 24 dB	4 dB
Cutoff	drone.cutoff	30 .. 16000 Hz	320 Hz
Resonance	drone.resonance	0 .. 1	0.3
Env Amount	drone.env_amount	0 .. 6 oct	0.5 oct
Filter Decay	drone.decay	15 .. 2000 ms	900 ms
Key Track	drone.keytrack	0 .. 1	0.3

Accent	drone.accent	0 .. 1	0
Amp Release	drone.amp_decay	5 .. 2000 ms	2000 ms
Glide	drone.glide	1 .. 500 ms	300 ms
Level	drone.level	-60 .. 6 dB	-10 dB
Pan	drone.pan	-1 .. 1	0
Echo Send	drone.echo	0 .. 1	0.1
Vibrato	drone.vibrato	0 .. 100 ct	0 ct
Vibrato Rate	drone.vibrato_rate	1 .. 10 Hz	5 Hz
Reverb Send	drone.reverb	0 .. 1	0.55
Auto Pan	drone.auto_pan	0 .. 1	0.35
Low Cut	drone.low_cut	10 .. 500 Hz	40 Hz
Distance	drone.distance	0 .. 1	0
Blend Send	drone.blend	0 .. 1	0

Tape Keys



A tape keyboard, synthesised: every key has its own strip of tape, with its own small detune, level and onset lag, played by one capstan whose speed wanders (wow and flutter) and sags a little under many keys. A tape runs out after eight seconds, as the instrument's did. Three sets: a choir of several singers through the formants of a sung vowel, strings, and flute. Age puts more of the machine's character into the sound; the tape set can change between pieces.

Spread widens the mono keyboard without colouring it: the side is the keyboard through a chain of all-passes whose phase turns at corners spread over the band, added on one side and taken away on the other, so the two sides differ in phase, not in tone, and their sum -- what a mono speaker plays -- is the keyboard alone. At the default 0.7 the correlation of the two sides is about 0.35: wide, with a middle the chords stand on.

TAPE

Control	Key	Range	Default
Tapes	tape.set	Choir, Strings, Flute	Choir
Vowel	tape.vowel	0 .. 1	0.2
Wow	tape.wow	0 .. 30 ct	6 ct
Flutter	tape.flutter	0 .. 10 ct	2 ct
Motor Load	tape.sag	0 .. 5 ct	1 ct
Tone	tape.tone	1500 .. 16000 Hz	7000 Hz
Age	tape.age	0 .. 1	0.5

Level	tape.level	-60 .. 6 dB	-8 dB
Pan	tape.pan	-1 .. 1	-0.1
Echo Send	tape.echo	0 .. 1	0.1
Reverb Send	tape.reverb	0 .. 1	0.5
Low Cut	tape.low_cut	10 .. 500 Hz	150 Hz
Distance	tape.distance	0 .. 1	0
Spread	tape.spread	0 .. 1	0.7
Blend Send	tape.blend	0 .. 1	0

Strings



A string machine of the seventies, after the example of Waldorf's Streichfett: top-octave generators divided down into five registers per key -- sawtooths at 16', 8' and 4' -- squares at 8' and 4' -- through an ensemble of bucket-brigade delays swept by a slow and a fast modulation. Keys glide from where they were when a chord is stolen, as the machine's did.

Registration morphs continuously through eight mixes of the registers: Violins, Violas, Cellos, Basses, Full, Hollow, Brass and Organ, each with its own brightness and all equally loud. Animate lets a slow sine move the registration by itself, up to three and a half mixes either way at the Animate Rate, so the blend of the voices wanders under a held chord; the composer's hands move the registration now and then as well. Octave Balance tilts the footages towards the low or the high ones. Ensemble Type chooses the chorus: Solina (three lines, a slow and a fast sweep), Chorus (two lines, one sweep) or Wide (three longer, deeper lines, spread wider). Phaser sends the machine through four swept all-passes, the classic swirl of strings through a phase shifter.

STRINGS

Control	Key	Range	Default
Crescendo	strings.attack	0.005 .. 3 s	0.35 s
Sustain	strings.release	0.05 .. 6 s	1.2 s
Octave Balance	strings.feet	0 .. 1	0.4
Tone	strings.tone	800 .. 12000 Hz	4500 Hz
Ensemble	strings.ensemble	0 .. 1	0.8

Level	strings.level	-60 .. 6 dB	-10 dB
Pan	strings.pan	-1 .. 1	0.1
Echo Send	strings.echo	0 .. 1	0.05
Reverb Send	strings.reverb	0 .. 1	0.45
Registration	strings.registration	0 .. 7	0
Animate	strings.animate	0 .. 1	0.3
Animate Rate	strings.animate_rate	0.01 .. 1 Hz	0.05 Hz
Ensemble Type	strings.ensemble_type	Solina, Chorus, Wide	Solina
Phaser	strings.phaser	0 .. 1	0
Low Cut	strings.low_cut	10 .. 500 Hz	200 Hz
Distance	strings.distance	0 .. 1	0
Blend Send	strings.blend	0 .. 1	0

Atmosphere



Wind (filtered noise in a moving band), slow sweeps, and bursts of sample-and-hold bleeps on the notes of the scale: the space at the start of a piece and in the bridges, and the stars over the builds.

Grains is a granular cloud: short tones, each a sine under a soft window of 40 to 250 ms, on the notes of the scale over the current root two to three octaves up, each at its own place in the stereo field, as many per second as Grain Density says. A piece has the cloud with the style's chance (the Style page); the hands then raise it in the atmosphere, the bridges and the coda, and take it away when the rows come in.

ATMOS

Control	Key	Range	Default
Wind	atmos.wind	-60 .. 0 dB	-60 dB
Wind Tone	atmos.wind_tone	150 .. 4000 Hz	700 Hz
Sweeps	atmos.sweeps	0 .. 6 /min	0 /min
Sweep Level	atmos.sweep_level	-60 .. 0 dB	-18 dB
Bleeps	atmos.bleeps	0 .. 12 /min	0 /min
Bleep Level	atmos.bleep_level	-60 .. 0 dB	-22 dB
Level	atmos.level	-60 .. 6 dB	-4 dB
Echo Send	atmos.echo	0 .. 1	0.2
Reverb Send	atmos.reverb	0 .. 1	0.6
Grains	atmos.grains	-60 .. 0 dB	-60 dB

Grain Density	atmos.grain_density	1 .. 40 /s	12 /s
Low Cut	atmos.low_cut	10 .. 500 Hz	120 Hz

Echo + Spring



A tape echo with a moving read head -- wow, flutter, saturation and loss of highs in the feedback -- synchronised to the tempo, with a ping-pong option; its feedback can run into self-oscillation, which the echo throw uses. Springs are built into it, as in the tape echoes of the time: they take the echo's send and return beside it. The composer keeps the bass and the pads out of the echo -- the bass dry, the tape keys and the strings in the hall only -- and the hands lean the feedback high in the quiet parts of a piece and low at the peak. Low Cut is a high pass inside the loop, so every repeat is thinner than the one before, as a distant sound is; Duck lowers the return (of both echoes) while the rows play. The repeats of both echoes sit at about 70 % of the stereo base rather than hard left and right, so the middle keeps a share of each and the ping-pong survives in mono.

Echo switches the same send to a bucket-brigade delay instead of the tape: an analogue delay of 4096 stages as in the pedals of the late seventies. Its clock runs slower the longer the delay is set, and its filters have to follow the clock, so the longer the echo, the darker its repeats; the clock wanders slowly (Wow sets how far), and the compander saturates and hisses faintly in the loop. Time, feedback, tone, drive and ping-pong work as on the tape.

Echo 2 is a second, cleaner tape echo on a send of its own (the rows' Echo 2 Send), with its own time, feedback, tone, ping-pong and return; its repeats go into the hall as the first echo's do. It lets each layer of the sequence have its own delay: the main sequence on the first echo's dotted eighths, the other counter rows on the second in eighths or quarter triplets (1/4T, which both echoes offer), as the composer draws it for each piece.

ECHO

Control	Key	Range	Default
Time	echo.time	1/16, 1/8, 3/16, 1/4, 3/8, 1/2, 1/4T	3/16
Feedback	echo.feedback	0 .. 1.1	0.45
Tone	echo.tone	800 .. 12000 Hz	3500 Hz
Wow	echo.wow	0 .. 4 ms	0.2 ms
Flutter	echo.flutter	0 .. 0.5 ms	0.02 ms
Tape Drive	echo.drive	0 .. 18 dB	4 dB
Ping-Pong	echo.pingpong	off / on	on
Return	echo.return	-60 .. 6 dB	-4 dB
Echo	echo.type	Tape, BBD	Tape
Low Cut	echo.low_cut	20 .. 1000 Hz	200 Hz
Duck	echo.duck	0 .. 12 dB	3 dB

SPRING

Control	Key	Range	Default
Decay	spring.decay	0.3 .. 8 s	2.2 s
Tone	spring.tone	1500 .. 9000 Hz	4500 Hz
Return	spring.return	-60 .. 12 dB	-6 dB

DELAY

Control	Key	Range	Default
Echo 2 Time	delay.time	1/16, 1/8, 3/16, 1/4, 3/8, 1/2, 1/4T	1/8
Echo 2 Feedback	delay.feedback	0 .. 1	0.3
Echo 2 Tone	delay.tone	800 .. 12000 Hz	5000 Hz
Echo 2 Ping-Pong	delay.pingpong	off / on	on
Echo 2 Return	delay.return	-60 .. 6 dB	-6 dB
Echo 2 Low Cut	delay.low_cut	20 .. 1000 Hz	300 Hz

Hall



A long, dark hall (an eight-line feedback delay network) with its size, decay, damping, pre-delay and the band it returns -- by default 250 Hz to 6.5 kHz, so the hall adds space without mud or glare. The composer lets its decay grow in the atmosphere, the bridges and the coda and shrink by 40 % at the peak, and sets the pre-delay to a 64th note at the piece's tempo (32 ms at 118 BPM), which keeps the dry sounds in front of their room. Duck lowers the hall's return by up to that many dB while the rows play, so the tail blooms in the gaps and steps back under the notes.

The blend room is a second, short plate (Blend Decay about a second, a few ms of pre-delay, 240 Hz to 7 kHz) on a send of its own (Blend Send on the rows, the lead, the drone, the tape keys, the strings and the drums). Into Hall sends a share of its return on into the hall, so the far room sounds like the near one going on, not like a second place -- the serial far space of the dark-ambient practice. The composer puts the second layer there - - the counter rows, the lead, the drums -- with only a little hall; the pads, the drone and the atmosphere stay in the hall, the bass stays dry; at the peak the feed into the hall drops to 5 %, where the far room would fill up. Both returns are ducked by the rows alike.

Room switches the same send to a plate instead: Dattorro's plate reverberator (1997) -- four diffusers and a tank of two halves that feed each other, each with a slowly moving all-pass, a damping low pass and the decay -- denser and brighter than the hall and without a room's early reflections, the sound of the plates on the records of the seventies. Decay, damping, pre-delay and the band work on it as well; its return is matched to the hall's level, so the switch changes the room, not the loudness.

REVERB

Control	Key	Range	Default
Size	reverb.size	0.3 .. 3	1.8
Decay	reverb.decay	0.3 .. 20 s	5.5 s
Damping	reverb.damping	0 .. 1	0.45
Pre-Delay	reverb.predelay	0 .. 200 ms	25 ms
Low Cut	reverb.lowcut	40 .. 500 Hz	250 Hz
High Cut	reverb.highcut	1000 .. 20000 Hz	6500 Hz
Return	reverb.return	-60 .. 12 dB	4 dB
Room	reverb.type	Hall, Plate	Hall
Duck	reverb.duck	0 .. 12 dB	2 dB

BLEND

Control	Key	Range	Default
Blend Decay	blend.decay	0.3 .. 4 s	1.2 s
Blend Pre-Delay	blend.predelay	0 .. 60 ms	8 ms
Blend Low Cut	blend.low_cut	60 .. 800 Hz	240 Hz
Blend High Cut	blend.high_cut	1500 .. 16000 Hz	7000 Hz
Blend Damping	blend.damping	0 .. 1	0.4
Blend Return	blend.return	-60 .. 6 dB	-2 dB
Into Hall	blend.into_hall	0 .. 0.5	0.15

Drums



A small analogue kit for the styles that have drums: kick, snare, hats, toms tuned to the root, rim and shaker, with patterns per style. The drums come late in a piece, from the second build to the end of the peak, and never before 45 % of the piece or after 90 %: the sequence is there before them and stays after them.

DRUMS

Control	Key	Range	Default
Kick Pitch	drums.kick_hz	35 .. 80 Hz	50 Hz
Decay	drums.decay	0.3 .. 2	1
Tone	drums.tone	0 .. 1	0.5
Level	drums.level	-60 .. 6 dB	-9 dB
Echo Send	drums.echo	0 .. 1	0.05
Reverb Send	drums.reverb	0 .. 1	0.2
Low Cut	drums.low_cut	10 .. 500 Hz	25 Hz
Blend Send	drums.blend	0 .. 1	0

Master



The master: its level, a gentle bus compressor that keeps the music's dynamics (a ratio of at most 1.5), and a true-peak limiter as protection at -1 dBTP. Before them the mix bus removes DC and subsonics under 20 Hz, keeps everything under 100 Hz in mono whatever its pan, and widens what lies above 300 Hz by Width (dB of side) -- never so far that the side comes within 3 dB of the mid, so the correlation stays positive and the mix survives in mono. Mono sums both sides after the limiter, to check the mix as a small speaker would play it. Soft Clip rounds the peaks above the ceiling into at most that many dB before the limiter (with anti-aliasing), so short ratchets do not make the limiter work; quieter material passes untouched. Sub Ceiling holds the band under 80 Hz on its own under that level (a Linkwitz-Riley split), so the sub never pushes the whole mix into the limiter. The composer moves the level along the form instead of compressing it -- the spaces 3 dB under the peak, the entry and the breakdown 1.5 dB -- and the width too: widest in the spaces, narrow where the dry bass comes in, growing again with the layers; while the lead plays, the tape keys and the strings step back 3 dB.

The styles are levelled to the loudness a streamed release is played at (EBU R128, measured with the offline renderer on ten-minute pieces): Cosmic, Melodic and Modern about -15.5 LUFS integrated, Doom and Drift about -19; the loudest three seconds around -12 LUFS, the true peak at -1 dBTP, the peak-to-loudness ratio 11 to 14 dB and the loudness range 8 to 13 LU -- the dynamics are the product. eph_render prints these figures for everything it renders, with the stereo correlation and how far the side lies under the mid. Below, the composer's settings -- the style, key, scale, tempo, lengths and whether the style chooses the tempo -- which the top bar also shows.

Album Form makes a concert an album: an interlude of three to five minutes between the long pieces -- no rows, only the drone, a slow chord every 16 to 32 bars on the strings or the tape keys, the atmosphere with its

grains, a longer hall and the open fifth at the end -- the piece where the concert's middle falls the darkest (Phrygian, darker voices, a row more at the peak), and the last one ethereal (Dorian, a longer hall, a row fewer, no drums, a quieter lead). Without it a concert is what it was.

Motion keeps the depth awake: a spectrum that does not change for twenty seconds loses its depth to the ear. The pads' levels move by a few tenths of a dB within a second or two; their hall sends, the strings' tone and the echo's feedback over 13 to 30 seconds; the atmosphere swells by 6 dB over 41 seconds -- periods that never lock into the sequence's cycles, so foreground and background do not breathe together. In the atmosphere, the bridges and the coda, where no sequence plays, a near event sounds now and then: a lead fragment of one to three notes, 20 to 90 seconds apart. Sub Solo leaves only what lies under 80 Hz, for checking the bottom of the mix; Mono and Sub Solo are for listening and change nothing that is exported unless they are on.

Exports (the plugin's and eph_render's) end with twenty seconds of the rooms after the last note, faded in over two seconds and out over the last ten, never linearly, so the last tail never breaks off. eph_render --archive writes an archive master besides: no limiter, 24 bit, the whole scaled to a true peak of -3 dBTP (it renders twice). The report adds the crest factor (true peak over RMS, 17 to 20 dB for the styles; 12 is the least that keeps the transients) and, with --stems, the correlation of each stem: the rows are mono points (1.0), the tape keys spread by all-pass decorrelation (about 0.35), the rooms enveloping (about 0.2).

OSC Cues sends what the score knows to a visualiser such as Kaleidoscope, over UDP at the OSC Port, to the machine named by the environment variable EPH_CUE_HOST (this machine without it): every beat with the tempo (/eph/beat), every section with its phase (/eph/phase), every new root (/eph/key) and every conjunction of the rows with the share of the running rows that meet (/eph/conjunction). Each cue leaves when its sound is heard, not when it is composed.

MASTER

Control	Key	Range	Default
Level	master.level	-60 .. 6 dB	0 dB
Compress	master.compress	0 .. 1	0.5
Ceiling	master.ceiling	-6 .. 0 dBTP	-1 dBTP
Width	master.width	-6 .. 6 dB	4.5 dB
Mono	master.mono	off / on	off
Cascade Duck	master.cascade	0 .. 6 dB	2.5 dB
Motion	master.motion	0 .. 1	1
Sub Solo	master.sub_solo	off / on	off
Soft Clip	master.clip	0 .. 2 dB	0.75 dB
Sub Ceiling	master.sub_ceiling	-18 .. 0 dBFS	-6 dBFS

COMPOSE

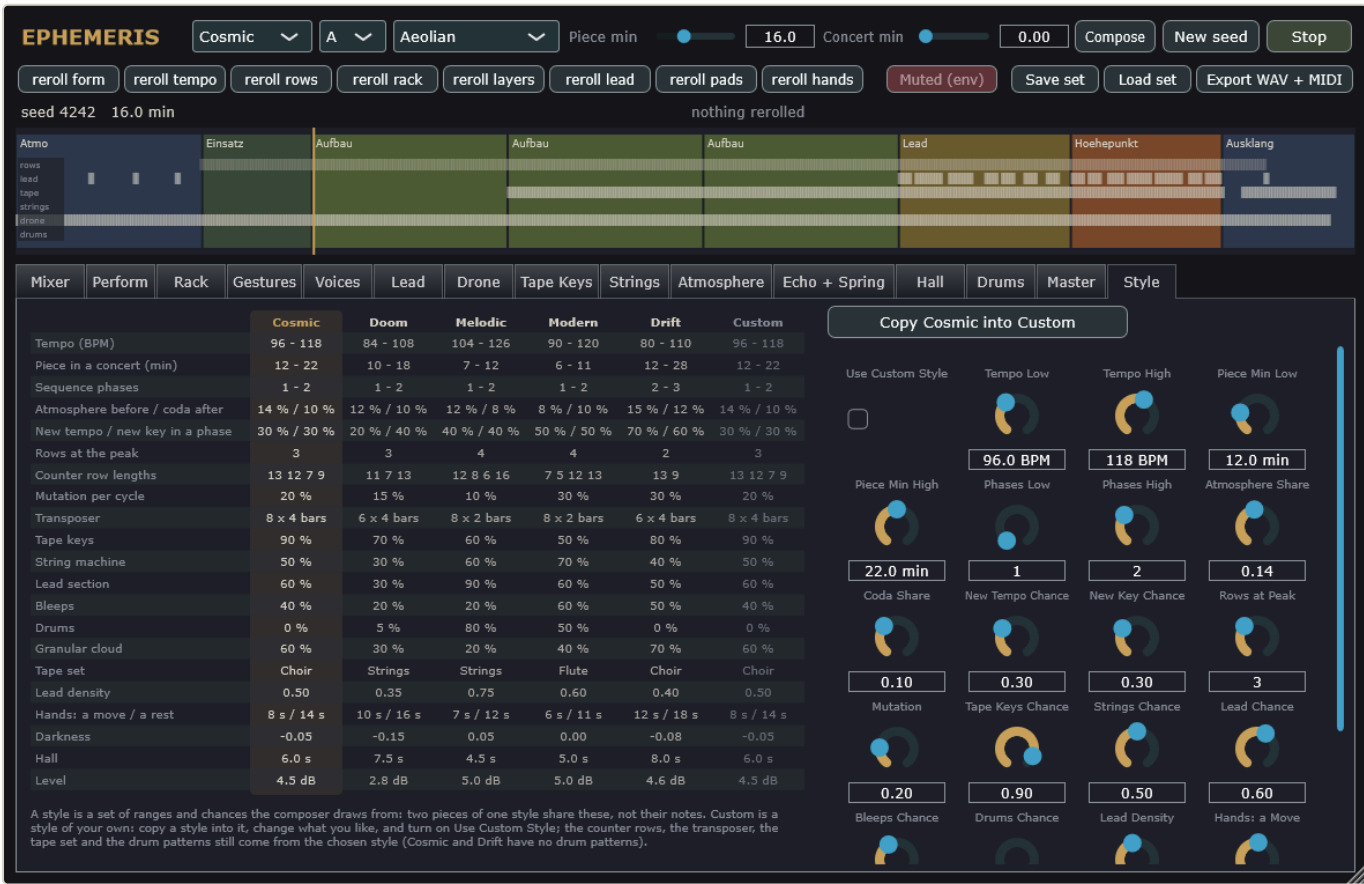
Control	Key	Range	Default
Tempo	compose.bpm	60 .. 160 BPM	118 BPM
Key	compose.key	C, C#, D, D#, E, F, F#, G, G#, A, A#, B	A

Scale	<code>compose.scale</code>	Aeolian, Dorian, Phrygian, Harmonic Minor, Minor Pentatonic, Mixolydian, Lydian, Locrian	Aeolian
Style	<code>compose.style</code>	Cosmic, Doom, Melodic, Modern, Drift	Cosmic
Piece Length	<code>compose.piece_minutes</code>	4 .. 40 min	16 min
Style Tempo	<code>compose.style_tempo</code>	off / on	on
Concert Length	<code>compose.concert_minutes</code>	0 .. 240 min	0 min
Morph To	<code>compose.morph_to</code>	None, Cosmic, Doom, Melodic, Modern, Drift	None
Concert Arc	<code>compose.concert_arc</code>	0 .. 1	0
Album Form	<code>compose.album</code>	off / on	off

CUE

Control	Key	Range	Default
OSC Cues	<code>cue.enabled</code>	off / on	off
OSC Port	<code>cue.port</code>	1024 .. 65535	9000

Style



The five style profiles side by side, the chosen one lit. A style is not a sound but a set of ranges and chances the composer draws from: the tempo, the length and the number of sequence phases, how much of the piece is atmosphere before the rows and after them, how often a later phase changes tempo or key, how many rows play at the peak and which lengths they are drawn from, how often a row mutates, how slowly the transposer moves, how often each layer comes in, how dense the lead is, how fast the hands move, how dark the voices are, how long the hall rings and how loud the piece is. Two pieces of one style share these, not their notes. The tempi and the levels were set after measuring reference recordings.

The sixth column is a style of your own. "Copy ... into Custom" takes the chosen style's numbers as a starting point; change what you like on the right, and turn on Use Custom Style: the composer then takes your numbers. The counter rows' lengths, the transposer, the tape set and the drum patterns still come from the chosen style -- Cosmic and Drift have no drum patterns, so there Drums Chance does nothing. Your style is saved with the set and the plugin's state, and a concert starts from it (and morphs from it).

CUSTOM

Control	Key	Range	Default
Use Custom Style	custom.use	off / on	off
Tempo Low	custom.bpm_low	60 .. 160 BPM	96 BPM
Tempo High	custom.bpm_high	60 .. 160 BPM	118 BPM
Piece Min Low	custom.minutes_low	4 .. 40 min	12 min
Piece Min High	custom.minutes_high	4 .. 40 min	22 min

Phases Low	custom.phases_low	1 .. 4	1
Phases High	custom.phases_high	1 .. 4	2
Atmosphere Share	custom.intro	0 .. 0.3	0.14
Coda Share	custom.coda	0 .. 0.3	0.1
New Tempo Chance	custom.new_tempo	0 .. 1	0.3
New Key Chance	custom.new_key	0 .. 1	0.3
Rows at Peak	custom.peak_rows	1 .. 7	3
Mutation	custom.mutation	0 .. 1	0.2
Tape Keys Chance	custom.tape	0 .. 1	0.9
Strings Chance	custom.strings	0 .. 1	0.5
Lead Chance	custom.lead	0 .. 1	0.6
Bleeps Chance	custom.bleeps	0 .. 1	0.4
Drums Chance	custom.drums	0 .. 1	0
Lead Density	custom.lead_density	0.1 .. 1	0.5
Hands: a Move	custom.hand_move	2 .. 30 s	8 s
Hands: a Rest	custom.hand_rest	4 .. 40 s	14 s
Darkness	custom.darkness	-0.5 .. 0.5	-0.05
Hall	custom.hall	1 .. 15 s	6 s
Level	custom.level	-12 .. 12 dB	4.5 dB